

Roll No. :

MID SEMESTER EXAMINATION MARCH 2025

Name of the Program: **B.Tech**

Semester: VI

Name of the Course: Advanced Machine Learning

Course Code: TCS 682

Time : 90 Minutes

Maximum Marks : 50

Note:

- i. Answer all the questions by choosing any one of the sub questions
- ii. Each questions carries 10 marks

Q1	(10 Marks)	CO1												
(a)	What is Expected Loss in decision theory, and how do you calculate it? What is the difference between generative models and discriminative models.													
	OR													
(b)	Define conditional probability and explain Bayes' Theorem with an example.	CO2												
Q2	(10 Marks)	CO2												
(a)	Using the k-nearest neighbors (k-NN) algorithm, classify the following point (3,4) using k=3 from the dataset: {(2, 3, 'Class A'), (5, 5, 'Class B'), (1, 2, 'Class A'), (6, 7, 'Class B')}. (Make the assumptions as per your choice.)													
	OR													
(b)	For a decision tree, compute the information gain for splitting on a feature (e.g., color) based on the following data. The dataset contains 10 instances, where 6 are class 'A' and 4 are class 'B'. After the split on color, 4 instances of 'A' and 1 of 'B' are in one branch, and 2 instances of 'A' and 3 of 'B' are in the other branch.													
Q3	(10 Marks)	CO2 ,3												
(a)	Given the following dataset: <table><tr><th>X (Feature)</th><th>Y (Target)</th></tr><tr><td>1</td><td>2</td></tr><tr><td>2</td><td>4</td></tr><tr><td>3</td><td>6</td></tr><tr><td>4</td><td>8</td></tr><tr><td>5</td><td>10</td></tr></table> i) Perform simple linear regression to find the best-fit line equation $Y=mX+b$, where m is the slope and b is the intercept. ii) Once you have the equation of the line, predict the value of Y when X=6. iii) Calculate the residuals for each data point and compute the Mean Squared Error (MSE).	X (Feature)	Y (Target)	1	2	2	4	3	6	4	8	5	10	
X (Feature)	Y (Target)													
1	2													
2	4													
3	6													
4	8													
5	10													
	OR													
(b)	What is the curse of dimensionality? Discuss how the curse of dimensionality affects the performance of nearest neighbor algorithms.	CO3												
Q4	(10 Marks)	CO1												

(a)	Define entropy in context of information theory. Calculate the entropy for a binary variable X with probabilities $P(X=0)=0.6$ and $P(X=1)=0.4$.	
	OR	
(b)	What is Lasso regression, and how does it help in feature selection?	
Q5	(10 Marks)	CO1
(a)	What is Kernel Density Estimation (KDE)?	CO2
	OR	
(b)	Explain the difference between parametric and non-parametric density estimation methods.	